

Hayden OUTLAW

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OBJECTIVE: To leverage a simple and creative approach to computing and modeling in order to drive innovative solutions and communicate knowledge.

EDUCATION

North Carolina State University — *PhD, Applied Mathematics* AUG 2024 – PRESENT

Goodnight Doctoral Fellow | Advised by Dr. Yeonjong Shin

Research in operator learning for dynamical systems, reduced-order modeling, and physics-informed neural networks.

North Carolina State University — *MS, Applied Mathematics* AUG 2024 – MAY 2026

Goodnight Fellow | GPA: 3.76/4.0 – Summa Cum Laude

Qualifying exams in Biomathematical Modeling, Numerical Analysis, and Partial Differential Equations

Selected Projects

- Applied deep neural networks and physics-informed modeling to computational fluid dynamics problems in operator space.
- Used encoder-decoder frameworks to identify and correct ionospheric interference in synthetic-aperture radar (SAR) imaging inversion problems.

Tulane University of Louisiana — *BS, Mathematics & Computer Science* AUG 2020 – MAY 2024

Minor in Philosophy | GPA: 3.93/4.0 | Advised by Dr. Daniel Bernstein

Awards: Crest Medal for Student Leadership; Terry Lawson Prize in Mathematics

Selected Projects

- Developed a retrieval-augmented generation (RAG) tool to answer natural-language queries over local city council meeting transcripts.
- Analyzed the effects of climate events on police and service call rates across groups, locations, and time.
- Built a training pipeline, visualizer, and query application for embedding-space control vectors in the Mistral-7B language model.

RESEARCH & PROFESSIONAL EXPERIENCE

North Carolina State University — *Mathematics Instructor* AUG 2024 – PRESENT

- Taught MA 131 and MA 141, introductory calculus courses at NC State.
- Developed and administered in-person and virtual courses, as well as one-on-one instruction and tutoring resources.

Vanderbilt University Medical Center — *Database Developer* MAY 2024 – AUG 2024

- Built a Java data-access framework with JPA and Hibernate for interacting with VUMC patient databases.
- Constructed an Angular front-end application for filtering, querying, and extracting data in a no-code, accessible manner.
- Collaborated with VUMC developers, engineers, and researchers in an Agile framework to improve research, patient confidentiality, and security workflows.

Tulane University, Department of Mathematics — *Undergraduate Research Assistant* AUG 2021 – DEC 2023

- Researched the graphical lasso covariance estimator and developed a novel algorithm that bypasses convex optimization using algebraic geometry.
- Numerically modeled a novel probability function, characterizing previously unknown behavior with high confidence and interpretability.
- Presented findings at the SIAM Conference on Applied Algebraic Geometry (AG23) at Eindhoven University of Technology, Netherlands, in July 2023.

National Center for Atmospheric Research (NCAR) — *Machine Learning Intern* MAY 2023 – AUG 2023

- Streamlined a distributed PyTorch neural network used to decompress cloud particle algorithms, improving performance and reducing computational load (SIParCS program, Computational and Information Systems Lab).
- Handled data management and processing, as well as parallel model training and evaluation, on the NSF NCAR Cheyenne and Derecho supercomputers.
- Collaborated with NOAA and NCAR developers, atmospheric scientists, and hardware engineers to deliver a practical, effective model framework.

Georgia Institute of Technology — *Biomathematics Researcher* MAY 2021 – AUG 2021

- Built an interactive, real-time agent-based model of a savanna ecosystem in NetLogo, covering agent behavior, atmospheric conditions, and reproduction rates, through the Southeast Center for Mathematics and Biology.

PUBLICATIONS & PRESENTATIONS

- H. Outlaw and D. I. Bernstein. *Maximum likelihood thresholds of Gaussian graphical models and graphical lasso*. arXiv preprint, 2024. [arXiv:2312.03145](#)
- **American Meteorological Society Annual Meeting** – Baltimore, MD, January 2024
 - **Tulane University Undergraduate Research Symposium** – New Orleans, LA, October 2023
 - **NCAR SIParCS Seminar** – Boulder, CO, August 2023 | [Recording](#) | [Poster](#)
 - **SIAM Conference on Applied Algebraic Geometry (AG23)** – Eindhoven, Netherlands, July 2023 | [Poster](#)

SERVICE

- NCSU Mathematics Directed Reading Program — *Graduate Student Mentor*** AUG 2024 – PRESENT
- *Fall 2026*, Intro to Analog Computing, advising Marcus O’Hara
 - *Spring 2025*, Classical Machine Learning Algorithms, advising Aryan Parikh
- Intercollegiate Math Modeling Challenge — *Referee*** 2025 – PRESENT
- Refereed student-developed models, data processing, and results presentation.

- Tulane University — *Supplemental Instructor*** JAN 2021 – MAY 2024
- Assisted in teaching undergraduate single-variable and multivariable calculus through the Supplemental Instruction program.
 - Created review materials, lectures, and resources to help students with assignments and exams, in coordination with faculty and graduate assistants.

SKILLS

- Mathematics:** Statistical Inference, Probability Theory, Machine Learning Theory, Stochastic Process Modeling, Natural Language Processing, Algorithmic Analysis, Number Theory, Abstract Algebra, Scientific Computing, PDEs, Numerical Analysis
- Programming:** Python (JAX, PyTorch), R, MATLAB, Java, C, C++, HTML/CSS, JavaScript, TypeScript, NetLogo, SQL and DBMS Software, HPC/Cloud Resource Management, Git/DVC, Linux/UNIX
- Languages:** Spanish (fluent), German (novice)

ABOUT ME

I am a student, programmer, and mathematician in Raleigh, North Carolina, currently pursuing a PhD in Applied Mathematics at North Carolina State University. My current focus is on deep-learning-powered computational physics, operator learning, and reduced-order modeling. In my free time, I like to scuba dive, run, volunteer, and see live music whenever I can.